

No. of Functional Features Included in the Solution

Date	2 July 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the Smart Lender project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S. No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Data Preprocessing & Cleaning	Handled missing values, dropped the non-predictive Loan_ID column, and prepared the raw dataset.	Data Pipeline	Done	0.5
2	Categorical Feature Encoding	Encoded categorical fields (Gender, Married, Dependents, Education, Self_Employed, Property_Area, Loan_Status) using Label Encoding.	Data Pipeline	Done	0.5
3	Multi-Model Training & Comparison	Trained and compared Decision Tree, Random Forest, KNN, and XGBoost for loan repayment/default prediction.	ML Engine	Done	1.0
4	Hyperparameter Tuning	Used GridSearchCV to tune Random Forest parameters via cross-validation.	ML Engine	Done	0.5
5	Best Model Selection & Evaluation	Selected XGBoost as best performer (94.7% training / 81.1% testing accuracy).	ML Engine	Done	0.5
6	Model Persistence	Serialized the trained model using joblib for reuse in the web app.	ML Engine	Done	0.5
7	Real-Time Prediction Web	Flask application that accepts applicant details and returns	Backend / Web App	Done	1.0

S. No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
	App	an instant prediction. Built and deployed.			
8	Applicant Input Form / UI	Web interface for submitting gender, marital status, education, employment, income, loan amount/term, credit history, and property area.	Frontend / UI	Done	0.5

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	1	Applicant input form for entering loan-related details
2	Backend / Logic	5	Data preprocessing, model training, hyperparameter tuning, model persistence, prediction logic
3	Database / Storage	0	Not used — dataset sourced from a static CSV file; no persistent database in the current build
4	API / Integration	1	Deployed Flask endpoint that receives form data and returns the prediction
5	Security / Authentication	0	Not implemented in the current version